

STG17 · DAY 4 · AFRICA AFTER DARK

Setting up a Google Earth Engine project

To access VIIRS night-time lights from the NTL notebook — step by step, with screenshots

Three things to know before you click



0 \$

Free for research, teaching and training. No billing account to create.



1 project

Mandatory since 13 November 2024: every Earth Engine call goes through a Google Cloud project.



≈ 15 min

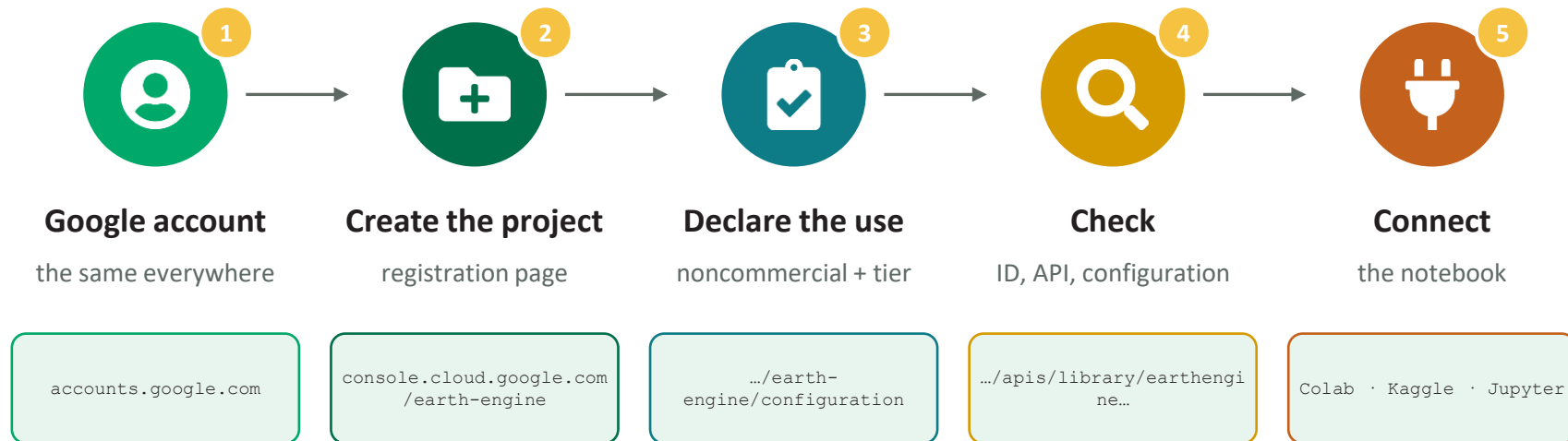
From project creation to the first VIIRS image in the notebook (workshop estimate).



Operational use: Google classes training and research as noncommercial, but steers any government agency using Earth Engine for operational purposes to the paid offer. Training staff this week: free. Producing the official indicator every month: to be clarified with Google.

Sources: Google Earth Engine — Transition to Cloud projects; Noncommercial use (earthengine.google.com/noncommercial).

Five steps, in this order



Steps 2 and 3: a single visit to the registration page is enough — it creates the project, enables the Earth Engine API and registers the use in one flow.

Source: Google Earth Engine — Earth Engine access (developers.google.com/earth-engine/guides/access).

01

CREATE AND REGISTER

Create and register the project

Which account to use, and where it all starts



Step 1 · One Google account, used everywhere



One account per person

Every participant who runs Earth Engine code needs their own Google identity (Gmail or Workspace).



The same account everywhere

Cloud Console, Colab and the authorisation window: one and the same account. It is the number-one cause of failure in the room.



Institutional account?

Some organisations forbid creating Cloud projects. In that case, ask IT to create the project and add you to it.

Step 2 · Create the project from the registration page

1 Open the page

console.cloud.google.com/earth-engine — signed in with the right account.

2 Create or choose

“Create a project”, or attach an existing one.

3 Name it

Free-form name; Google derives a unique, permanent ID. E.g. ntl-stg17 (a suffix is added if taken).

4 Confirm

The flow enables the Earth Engine API and moves on to the use declaration.

console.cloud.google.com/earth-engine

Register a project for Earth Engine

☒ Create a project ☐ Use an existing project

Project name

NTL-STG17

Project ID — permanent

ntl-stg17

Continue

SCHEMATIC — the real screen may differ

Source: Google Earth Engine — Earth Engine access, “Create a project”.

Step 3 · Declare noncommercial use

The questionnaire, mandatory since 26 September 2025, confirms the project's eligibility for free access. It must be revalidated periodically.

Noncommercial — free

- A training workshop such as STG17
- University research and teaching
- A nonprofit organisation, for noncommercial activities
- A PhD thesis on natural resources

Commercial — paid

- Selling a service built on Earth Engine
- Internal R&D for a commercial product
- Promotional white papers for a product
- A government agency using Earth Engine for operational purposes



How to answer: for this workshop, the declared use is training. Answer honestly: a project reclassified as commercial loses free access.

Sources: earthengine.google.com/noncommercial; Earth Engine access, “Configuring noncommercial access”.

Step 3 (continued) · Choose a quota tier

RECOMMENDED

Community

Default tier

No billing account. Enough for this notebook.

Contributor

Higher quota

Requires an attached billing account — not charged for Earth Engine under noncommercial use.

Partner

Highest quota

Its own eligibility criteria, for Google's partners.



Caution: if you attach a billing account (Contributor tier), do not enable any other Google Cloud API on this project: Earth Engine stays free, but other services can be charged. The tier can be changed at any time under Configuration.

Source: Google Earth Engine — Noncommercial tiers (developers.google.com/earth-engine/guides/noncommercial_tiers).

02

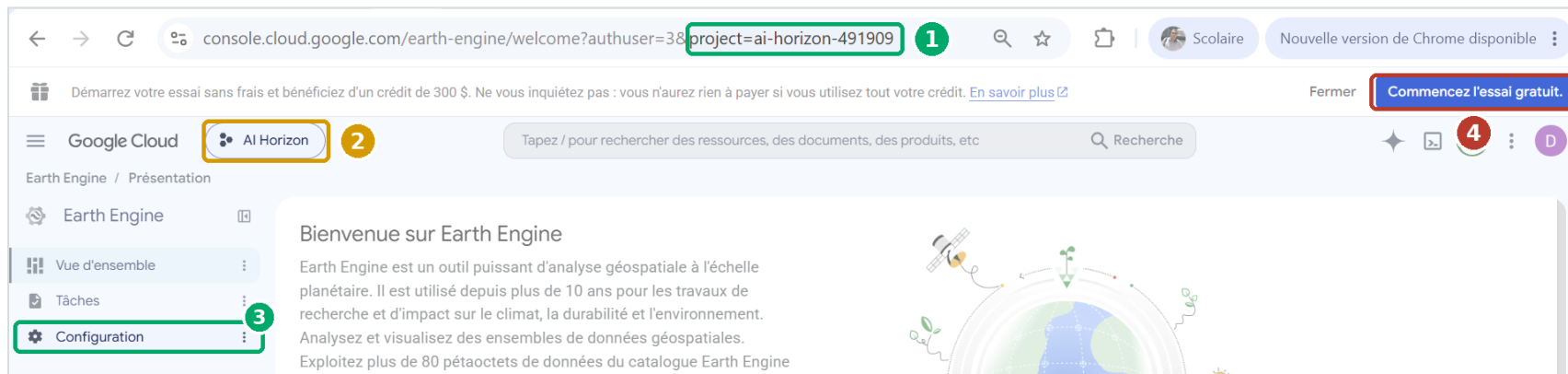
CHECK IN THE CONSOLE

Check in the console

Is your project really ready?



The Earth Engine console: four areas to spot



1 The project ID

in the address, after project=: this is what the notebook asks for.

2 The project name

“AI Horizon”: a free label, not the ID.

3 Configuration

registration, questionnaire and quota tier.

4 \$300 free trial

unrelated to Earth Engine. Do not click.

ID to enter in the notebook: **ai-horizon-491909** — *specific to this account; each participant has their own.*

Real screenshot · Google Cloud console, Earth Engine home page of the workshop project (French interface).

Name ≠ ID: do not mix them up

PROJECT NAME



Google Cloud



AI Horizon

Readable, editable, not necessarily unique. Earth Engine does not accept it.

PROJECT ID

ai-horizon-491909

Lowercase, digits, hyphens. Unique across Google Cloud. Permanent. This is what you type.

THREE WAYS TO FIND IT

1

Console address

the project= parameter in the URL

2

Project selector

click the name: the window shows name and ID

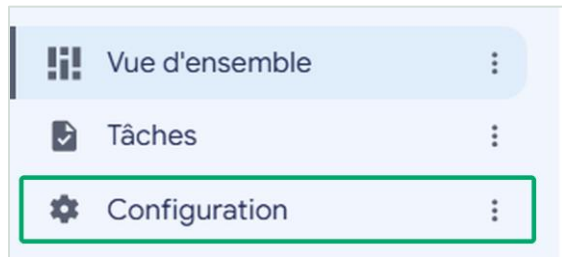
3

Code Editor Earth Engine

profile menu, top right

Real screenshot (project selector, Google Cloud console).

Check the registration and the API



Real screenshot · Earth Engine menu.

UNDER “CONFIGURATION”, YOU SHOULD SEE

- ✓ Project registered — noncommercial use
- ✓ Eligibility questionnaire completed
- ✓ Quota tier selected



Registration incomplete?

Resume the flow at this address, replacing with your ID:

`code.earthengine.google.com/register?project=YOUR-ID`



Is the Earth Engine API enabled?

Normally yes if you went through the registration page. Otherwise, open this page, check the project selected at the top, then click “Enable”:

`console.cloud.google.com/apis/library/earthengine.googleapis.com`

Sources: Earth Engine access; Authentication and Initialization, Troubleshooting section.

03

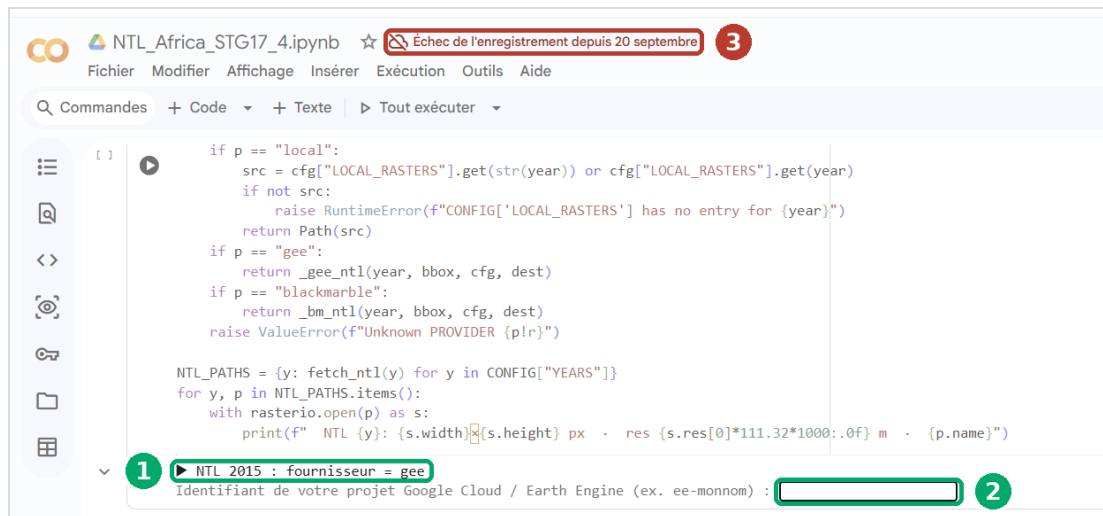
CONNECT THE NOTEBOOK

Connect the notebook

The last mile: from the ID to the first image



Answering the notebook prompt in Colab



1 The “gee” provider

the notebook starts the real acquisition of 2015.

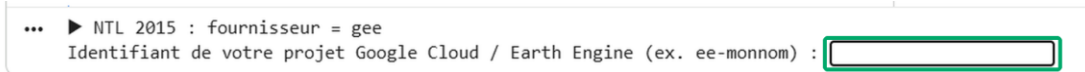
2 The input box

type the ID — no quotes or spaces — then Enter.

3 Save failure

the file is no longer saved. Close the other tabs, then File ▶ Save a copy in Drive.

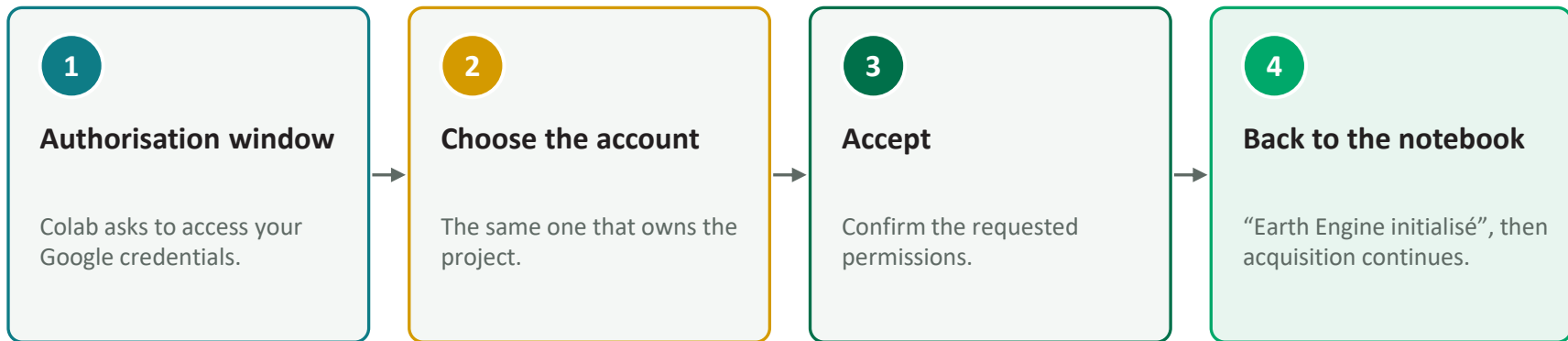
ZOOMED IN



Type: **ai-horizon-491909** ↵

Real screenshot · Google Colab, cell 4.2 of the NTL_Africa_STG17 notebook (cropped, French interface).

Authorising access: what appears next



```
► NTL 2015 : fournisseur = gee  
Identifiant ... : ai-horizon-491909  
! Authentification Earth Engine requise...  
✓ Earth Engine initialisé
```

Nothing opens?

Allow pop-ups for `colab.research.google.com`, then rerun the cell.

Wrong account chosen?

Run `ee.Authenticate(force=True)`, then rerun the cell.

Expected notebook output (reconstructed; the notebook prints in French). Source: Authentication and Initialization, “colab” mode.

The 30-second test, before “Run all”

```
import ee
ee.Authenticate()
ee.Initialize(project="ai-horizon-491909")

viirs = ee.ImageCollection("NOAA/VIIRS/DNB/ANNUAL_V22")
print(viirs.size().getInfo())
print(viirs.first().date().format("YYYY-MM-dd").getInfo())
```



It works

A positive integer, then a date. Project, API, permissions and authentication are in order.



It fails

The error message points to the cause. Recheck the account, the ID, the registration and the API.

Replace ai-horizon-491909 with your own project ID.

Standard calling code: ee.Authenticate() then ee.Initialize(project=...) — Authentication and Initialization.

Night-time lights available in Earth Engine

Earth Engine collection	Time step	Period	Use in the notebook
NOAA/VIIRS/DNB/ANNUAL_V22	Annual	2012 → 2024	CONFIG["YEARS"] — scenarios A to E and G
NOAA/VIIRS/DNB/MONTHLY_V1/VCMSLCFG	Monthly, stray-light corrected	Jan 2014 → Mar 2025	CONFIG["MONTHLY_YEARS"] — scenario F
NOAA/VIIRS/DNB/MONTHLY_V1/VMCMCFG	Monthly	Apr 2012 → Mar 2025	Alternative, not used by default
NASA/VIIRS/002/VNP46A2	Daily (Black Marble, BRDF-corrected)	see the catalogue	Not used — for going further



Practical consequence: the monthly series stops in March 2025 in the catalogue: keep MONTHLY_YEARS up to 2024 for complete years. Check the bounds before every run.

Resolution: 463.83 m for the NOAA/EOG products · Public-domain data, no restriction on use (Colorado School of Mines).

Source: Earth Engine Data Catalog — pages NOAA_VIIRS_DNB_ANNUAL_V22, MONTHLY_V1/VCMSLCFG, MONTHLY_V1/VMCMCFG; release notes (VNP46A2).

The checklist, before “Run all”

☐ One Google account, the same everywhere

☐ Project created from the registration page

☐ Noncommercial questionnaire completed

☐ Quota tier chosen — Community is enough

☐ Earth Engine API enabled

☐ ID noted down — not the name

☐ 30-second test passed

☐ Notebook saved in Drive

Tick in order: each box assumes the previous ones.

Sources

Earth Engine access

developers.google.com/earth-engine/guides/access

Authentication and Initialization

developers.google.com/earth-engine/guides/auth

Control access to resources (IAM roles)

developers.google.com/earth-engine/guides/access_control

Noncommercial tiers

developers.google.com/earth-engine/guides/noncommercial_tiers

Transition to Cloud projects

developers.google.com/earth-engine/guides/transition_to_cloud_projects

Noncommercial use — eligibility

earthengine.google.com/noncommercial

Catalogue: VIIRS DNB Annual V2.2

developers.google.com/earth-engine/datasets/catalog/NOAA_VIIRS_DNB_ANNUAL_V22

Catalogue: VIIRS DNB Monthly VCMSLCFG

developers.google.com/earth-engine/datasets/catalog/NOAA_VIIRS_DNB_MONTHLY_V1_VCMSLCFG

Documentation consulted in September 2026. Screenshots come from a real workshop project (French interface); schematics are indicative.

READY TO LAUNCH

Your project is ready. Lights on.

Type the ID, press Enter, authorise access — then let the notebook work through to the dashboard.

